

Lab4: Clustering (and PCA)

STAT 4610/5610

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Emery Individual Section

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
Attitude_clean <- read.csv("Attitude_clean.csv")
X <- dplyr::select(Attitude_clean, -Group)
X_scaled <- scale(X)

group_labels <- Attitude_clean$Group
```

Part 1. K-means Clustering

```
set.seed(123)
km.out <- kmeans(X_scaled, centers = 3, nstart = 20)

# Compare clusters to known groups
table(Cluster = km.out$cluster, Group = group_labels)
```

	Group					
Cluster	IndoPost24	IndoPre25	IndoProf24	SLSp25	SLSp26	StatCollab
1	23	28	17	8	8	25
2	5	15	9	0	0	0
3	2	12	10	3	13	12

Part 2. Comparison

```
# Compare clusters to known groups
table(Cluster = km.out$cluster, Group = group_labels)
```

	Group					
Cluster	IndoPost24	IndoPre25	IndoProf24	SLSp25	SLSp26	StatCollab
1	23	28	17	8	8	25
2	5	15	9	0	0	0
3	2	12	10	3	13	12

Using K-means clustering with K=3 and 20 random starts, the three clusters correspond somewhat with the 6 groups . Cluster 2 is the most distinct, containing only Indonesian respondents (5 IndoPost24, 15 IndoPre25, and 9 IndoProf24) and no American students. This suggests Indonesian respondents share a meaningfully different attitude profile compared to their American counterparts. Cluster 1 is the largest and most mixed, capturing some of every group, while Cluster 3 pulls in mostly American students (13 SLSp26 and 12 StatCollab) alongside some Indonesian students. Overall, K=3 reveals a meaningful distinction between Indonesian and American respondents, but a higher K would likely be needed to further differentiate within those two populations.

Part 3. Clustering Attitudes

```
# Transpose so attitudes are the observations
X_t <- t(X_scaled)

set.seed(123)
km.att <- kmeans(X_t, centers = 3, nstart = 20)

# See which attitudes fall in which cluster
data.frame(Attitude = rownames(X_t),
           Cluster = km.att$cluster) |>
  arrange(Cluster) |>
  split(~Cluster) |>
  print()
```

\$`1`

	Attitude	Cluster
A1.freq	A1.freq	1
A3.freq	A3.freq	1
A4.freq	A4.freq	1
A5.freq	A5.freq	1
A1.fav	A1.fav	1
A3.fav	A3.fav	1
A4.fav	A4.fav	1
A5.fav	A5.fav	1
A11.freq	A11.freq	1
A15.freq	A15.freq	1
A11.fav	A11.fav	1
A15.fav	A15.fav	1
A18.freq	A18.freq	1
A19.freq	A19.freq	1
A20.freq	A20.freq	1
A18.fav	A18.fav	1
A19.fav	A19.fav	1
A20.fav	A20.fav	1

\$`2`

	Attitude	Cluster
A2.freq	A2.freq	2
A2.fav	A2.fav	2
A8.freq	A8.freq	2
A9.freq	A9.freq	2
A10.freq	A10.freq	2
A8.fav	A8.fav	2
A9.fav	A9.fav	2
A10.fav	A10.fav	2
A22.freq	A22.freq	2
A24.freq	A24.freq	2
A22.fav	A22.fav	2
A24.fav	A24.fav	2

\$`3`

	Attitude	Cluster
A6.freq	A6.freq	3
A7.freq	A7.freq	3
A6.fav	A6.fav	3
A7.fav	A7.fav	3
A12.freq	A12.freq	3

A13.freq	A13.freq	3
A14.freq	A14.freq	3
A12.fav	A12.fav	3
A13.fav	A13.fav	3
A14.fav	A14.fav	3
A16.freq	A16.freq	3
A17.freq	A17.freq	3
A16.fav	A16.fav	3
A17.fav	A17.fav	3
A21.freq	A21.freq	3
A23.freq	A23.freq	3
A25.freq	A25.freq	3
A21.fav	A21.fav	3
A23.fav	A23.fav	3
A25.fav	A25.fav	3

For every attitude, its freq and fav versions land in the same cluster. This suggests that how often respondents reported having an attitude is closely tied to how they believed it would impact collaboration. Cluster 1 contains attitudes A1, A3, A4, A5, A11, A15, A18, A19, and A20; Cluster 2 contains A2, A8, A9, A10, A22, and A24; and Cluster 3 contains A6, A7, A12-A14, A16, A17, and A21-A25. Without the actual attitude descriptions it is difficult to interpret what substantively distinguishes the three clusters, but the consistent freq/fav pairing suggests a strong relationship between frequency and perceived impact across all attitudes.

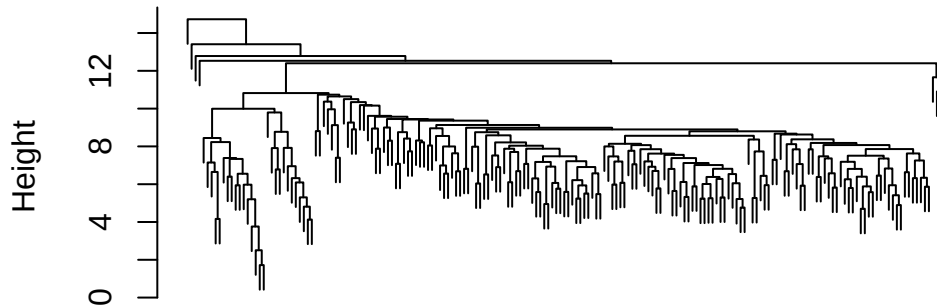
Part 4. Hierarchical Clustering - Average Linkage

```
# Plot both dendrograms to visually choose K
# Euclidean distance
hc.avg.euc <- hclust(dist(X_scaled, method = "euclidean"), method = "average")

# Manhattan distance
hc.avg.man <- hclust(dist(X_scaled, method = "manhattan"), method = "average")

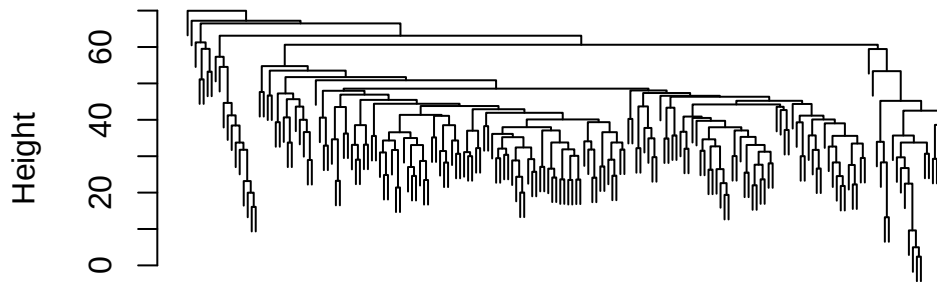
plot(hc.avg.euc, main = "Average Linkage, Euclidean", xlab = "", sub = "", labels = FALSE)
```

Average Linkage, Euclidean



```
plot(hc.avg.man, main = "Average Linkage, Manhattan", xlab = "", sub = "", labels = FALSE)
```

Average Linkage, Manhattan



```
best.k <- 3

cut.euc <- cutree(hc.avg.euc, k = best.k)
cut.man <- cutree(hc.avg.man, k = best.k)

table(Euclidean = cut.euc, Group = group_labels)
```

	Group					
Euclidean	IndoPost24	IndoPre25	IndoProf24	SLSp25	SLSp26	StatCollab
1	30	55	34	11	21	37
2	0	0	1	0	0	0
3	0	0	1	0	0	0

```
table(Manhattan = cut.man, Group = group_labels)
```

	Group					
Manhattan	IndoPost24	IndoPre25	IndoProf24	SLSp25	SLSp26	StatCollab
1	30	55	35	11	21	36
2	0	0	0	0	0	1
3	0	0	1	0	0	0

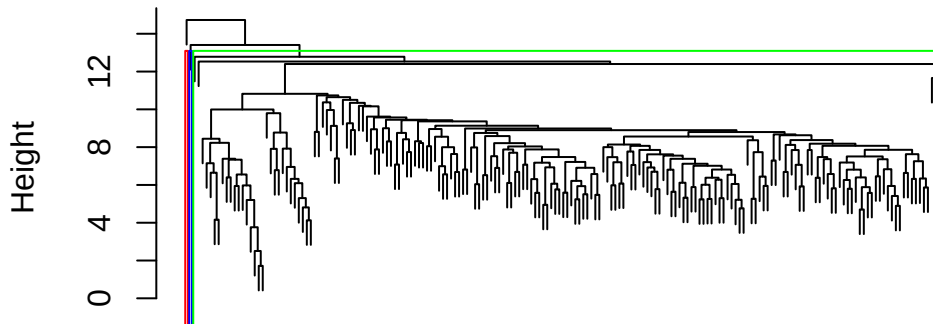
Both Euclidean and Manhattan distance gave identical results under average linkage, so Euclidean was kept as the default. K=3 was chosen to match the K-means analysis and allow for direct comparison, along with the fact that it had one of the largest height gaps on the dendrogram. However, regardless of K chosen, average linkage placed nearly all respondents into one cluster with only a single outlier separated out. This is a known issue with average linkage. Complete linkage was tested and confirmed to give much more balanced clusters, so the poor result here is due to the linkage method rather than the data or distance measure.

Part 5. Best Dendrogram

```
plot(hc.avg.euc,
     main = "Hierarchical Clustering of Respondents (Average, Euclidean)",
     xlab = "", sub = "",
     labels = FALSE)

rect.hclust(hc.avg.euc, k = 3, border = c("red", "blue", "green"))
```

Hierarchical Clustering of Respondents (Average, Euclidean)



The dendrogram confirms what the table showed. Average linkage with K=3 places nearly all respondents into a single large cluster, with two individual IndoProf24 respondents split off as their own clusters. This does not correspond meaningfully to the six known groups at all. Compared to K-means, which at least identified a distinct Indonesian cluster, hierarchical clustering with average linkage fails to recover any meaningful group structure in this data. This reinforces that average linkage is a poor choice for this dataset, because average linkage

tends to snowball in high-dimensional data, absorbing points one by one into one giant cluster rather than forming distinct groups.

Part 6. Cluster Attitudes

```
# Correlation-based distance
att.dist.corr <- as.dist(1 - cor(t(X_t)))
hc.att.comp.corr <- hclust(att.dist.corr, method = "complete")

# Euclidean for comparison
hc.att.comp.euc <- hclust(dist(X_t, method = "euclidean"), method = "complete")

# Cut at K=3
att.cut.corr <- cutree(hc.att.comp.corr, k = 3)
att.cut.euc <- cutree(hc.att.comp.euc, k = 3)

# Correlation distance
data.frame(Attitude = rownames(X_t), Cluster = att.cut.corr) |>
  arrange(Cluster) |>
  split(~Cluster) |>
  print()
```

```
$`1`
      Attitude Cluster
A1.freq  A1.freq      1
A3.freq  A3.freq      1
A4.freq  A4.freq      1
A1.fav   A1.fav      1
A3.fav   A3.fav      1
A4.fav   A4.fav      1
A8.freq  A8.freq      1
A8.fav   A8.fav      1
A11.freq A11.freq     1
A15.freq A15.freq     1
A11.fav  A11.fav     1
A15.fav  A15.fav     1
A18.freq A18.freq     1
A19.freq A19.freq     1
A20.freq A20.freq     1
A18.fav  A18.fav     1
A19.fav  A19.fav     1
```

A20.fav	A20.fav	1
---------	---------	---

\$`2`

	Attitude	Cluster
A2.freq	A2.freq	2
A5.freq	A5.freq	2
A2.fav	A2.fav	2
A5.fav	A5.fav	2
A9.freq	A9.freq	2
A10.freq	A10.freq	2
A9.fav	A9.fav	2
A10.fav	A10.fav	2
A22.freq	A22.freq	2
A24.freq	A24.freq	2
A22.fav	A22.fav	2
A24.fav	A24.fav	2

\$`3`

	Attitude	Cluster
A6.freq	A6.freq	3
A7.freq	A7.freq	3
A6.fav	A6.fav	3
A7.fav	A7.fav	3
A12.freq	A12.freq	3
A13.freq	A13.freq	3
A14.freq	A14.freq	3
A12.fav	A12.fav	3
A13.fav	A13.fav	3
A14.fav	A14.fav	3
A16.freq	A16.freq	3
A17.freq	A17.freq	3
A16.fav	A16.fav	3
A17.fav	A17.fav	3
A21.freq	A21.freq	3
A23.freq	A23.freq	3
A25.freq	A25.freq	3
A21.fav	A21.fav	3
A23.fav	A23.fav	3
A25.fav	A25.fav	3

```
# Count of attitudes per cluster  
table(Cluster = att.cut.cor)
```


Cluster		
1	2	3
18	12	20

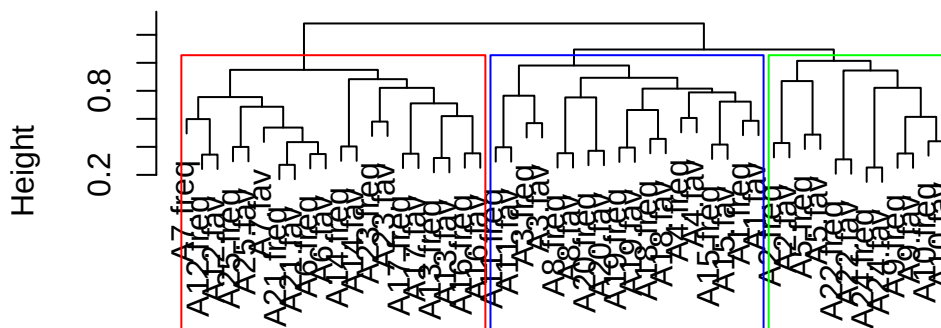
Correlation distance was used because we care about which attitudes move together across respondents, not their absolute values. Complete linkage was used instead of average linkage, which was shown earlier to chain badly on this dataset. The resulting three clusters are reasonably balanced with 18, 12, and 20 attitudes respectively. As with K-means, every attitude's freq and fav versions land in the same cluster, reinforcing that frequency and perceived impact are closely related.

Part 7. Best Dendrogram

```
plot(hc.att.comp.corr,
     main = "Hierarchical Clustering of Attitudes (Complete, Correlation)",
     xlab = "", sub = "")

rect.hclust(hc.att.comp.corr, k = 3, border = c("red", "blue", "green"))
```

Hierarchical Clustering of Attitudes (Complete, Correlatio



The dendrogram shows three decently balanced clusters with clear separation, suggesting complete linkage with correlation distance was a good fit for this data. The results closely mirror the K-means attitude clusters from Part 3 with the same attitudes tending to group together across both methods. This is strong evidence that these clusters reflect genuine structure in the data rather than an artifact of the algorithm chosen. As with K-means, every attitude's freq and fav versions land in the same cluster, reinforcing that frequency and perceived impact are consistently tied together regardless of the clustering method used.